Sales Data Report

Power BI report of online sales data

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Overview

The task here is to automate, clean and analyze a dirty sample invoice data supplied by a client.

Datasource https://foresightbi.com.ng/wp-content/uploads/2020/05/1.-Badly-Structured-Sales-Data-1.xlsx

Data was cleaned using power query and analyzed with Power BI

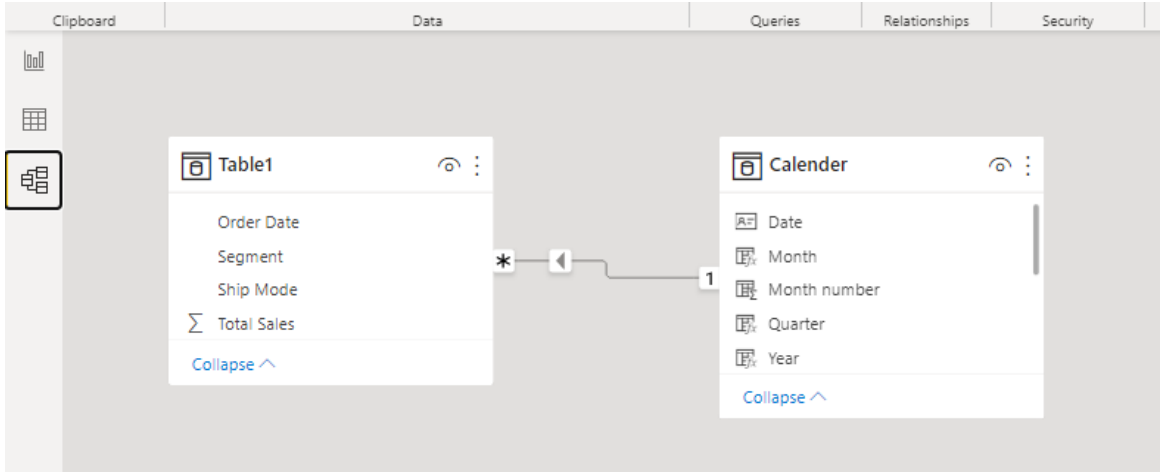
The raw Dataset is on the left side; The end results of cleaning is on the right

Segment	Consumer	Corporate	Home Office	Consumer	Corporate	Home Office	Consumer	Corporate	Home Office	Consumer	Corporate	Home Office
Order Date												
14-Mar-13												91.056
16-Dec-13							129.44					
2-Jun-13										605.47		
21-Oct-13											788.86	
27-Aug-13							13.36					
28-Nov-13								542.34				
31-Mar-13											1.869	
21-Nov-13											865.5	
1-Nov-13											1044.44	
5-Apr-13								4251.92				
5-Jul-13		242.546										
15-Jan-13	149.95											
2-Dec-13											21.19	
19-Mar-13		590.762										
27-Jun-13										616.14		
6-Jan-13		12.78										
14-May-13											310.88	
12-Dec-13										23.472		
29-Apr-13											661.504	
24-Oct-13												10.368

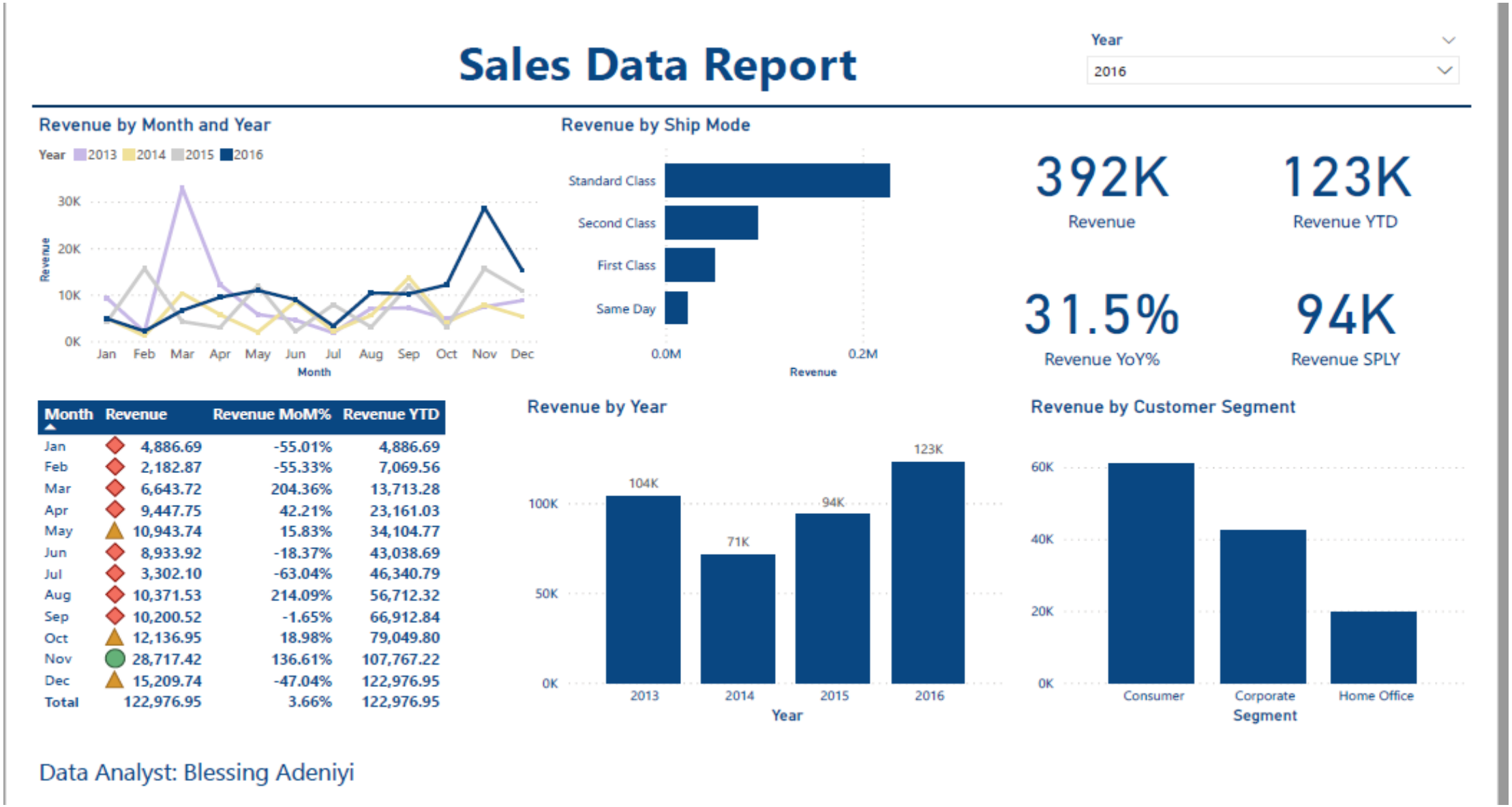
Ship Mode	Segment	Order Date	Total Sales
First Class	Consumer	15/01/2013	149.95
First Class	Consumer	15/08/2013	243.6
First Class	Consumer	24/12/2013	9.568
First Class	Consumer	07/04/2013	8.96
First Class	Consumer	19/05/2013	34.2
First Class	Consumer	05/09/2013	31.984
First Class	Consumer	12/08/2013	286.65
First Class	Consumer	05/07/2013	514.03
First Class	Consumer	30/04/2013	1000.95
First Class	Consumer	23/03/2013	9.912
First Class	Consumer	30/12/2013	39.128
First Class	Consumer	18/04/2014	106.5
First Class	Consumer	21/11/2014	18.176
First Class	Consumer	23/12/2014	194.32
First Class	Consumer	23/03/2014	59.48
First Class	Consumer	30/10/2014	182.91
First Class	Consumer	16/04/2014	2298.9
First Class	Consumer	02/11/2014	197.72
First Class	Consumer	30/11/2014	440.144
First Class	Consumer	22/11/2014	32.985
First Class	Consumer	11/12/2014	196.62

Dataset was analyzed and cleaned using Power BI

Data was imported and first transformed in Power query, the data has four rows as seen on slide 3 i.e. Ship mode, Segment, Order date and Sales. Using DAX, Additional columns were created on the date table and quick measure adopted to calculate the necessary variables from the revenue such as Revenue Year to Date, Revenue Month over Month Variance, Revenue Month to Date, Revenue Same Period Last Year etc. The invoice provided focused on a period of four calendar years (2013-2016) and the analysis concentrated on the peak period (2016), establishing Sales relationship between same and other years. Data was simply modelled by establishing a relationship between the dimension table and the Date table after which data was analyzed and visualized to draw insight.



Data Processing and Visualization



Key Insights

- Total Revenue generated over 4years was $392,000. 2016 had $123,000 with a Year over year growth of 31.5% compared to $94,000 generated in the previous year.
- In the customer segments, consumers had the highest sales report recording $60,000 which is about 50% of the total revenue.
- 2014 had a very low revenue compared to other years. It is necessary to probe further and find out why.
- Revenue was at its peak in November 2016 hitting $28,717 with a whooping 136.6% Month over Month percent increase. The increasing pattern was constant over the last 2 years.
- By Ship mode, Standard class had the highest revenue compared to First class, Second class and same day.

Conclusions and Recommendations

- Majority of the company's revenue was generated from standard class category of ship mode.
- Majority of the customers that purchased the products belongs to the consumers in the customer segments.
- Though the performance in 2014 was low compared to other years, there's a growing trend in the revenue generated over time with a tremendous year over year growth. The business is promising.
- Effective Sales strategy should be adopted to drive sales among the Corporate and Home Office segments.
- Customers should be engaged to obtain feedback on why they prefer Standard class of Shipping.
- Customer engagement and loyalty to the company is key.